

Qatar-1b

R-2574

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_110929 · created 2026-08-27T00:11:06+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2455832.23652 +/- 0.00038 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.099 +/- 0.036
Transit depth (Rp/Rs)^2	0.98 +/- 0.7 [%]
Orbital Inclination (inc)	86.2 +/- 2.62
Airmass coefficient 1 (a1)	0.1416 +/- 0.0015
Airmass coefficient 2 (a2)	1.62 +/- 0.41
Scatter in the residuals of the lightcurve fit is	1.06 %
Best Comparison Star	#1 - [527, 366]
Optimal Aperture	3.46
Optimal Annulus	9.35
Transit Duration (day)	0.0296 +/- 0.0058

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.099 $\pm$ 0.036 vs 0.1463 (deviation 0.95 σ)
Duration fit vs published	0.0296 d vs 0.0692 d (deviation 4.94 σ)
Mid-time O-C	-2044.4 min
Depth SNR	2.0
Residual scatter	1.06 %

Light curve

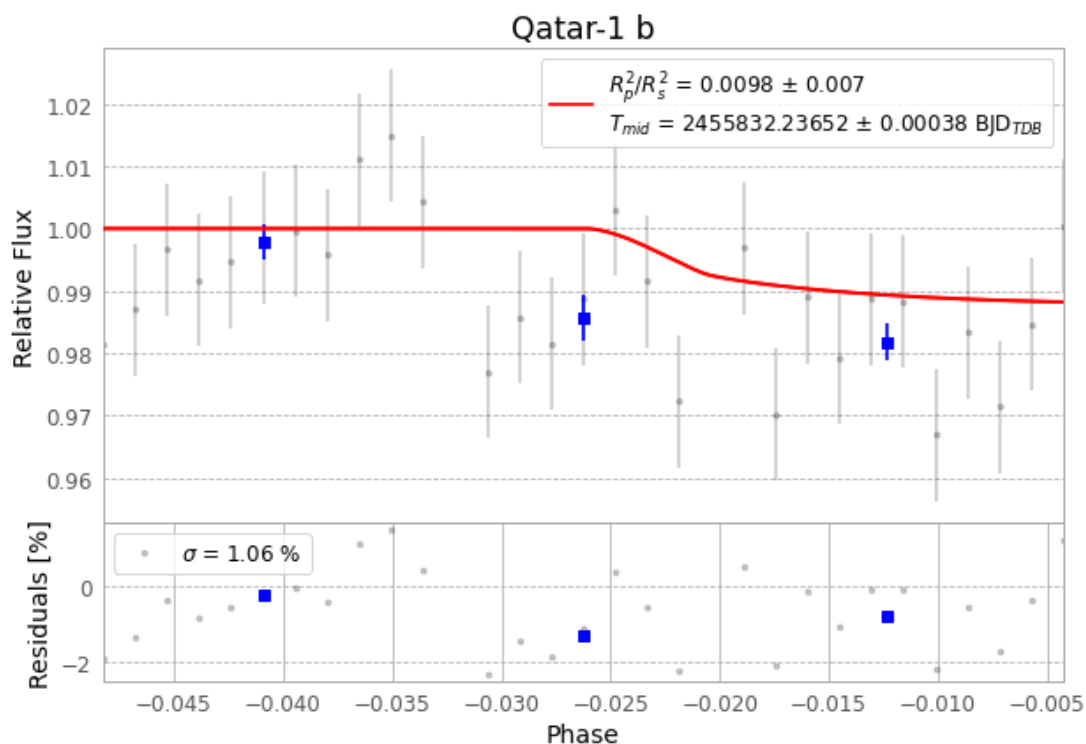


Figure 1 — FinalLightCurve_Qatar-1 b_2011-09-28.png (SHA-256 5a8c424d4ba0894a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [414, 288], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0cc24393f77d9509...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

31 FITS frames (20 MB), Qatar-1110929020312.FITS → Qatar-1110929033312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-110929.log (42e9da975294...), FinalLightCurve_Qatar-1 b_2011-09-28.png (5a8c424d4ba0...), FinalLightCurve_Qatar-1 b_2011-09-28.csv (bd8136e2c246...)

Compute resources

Wall time 107.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-27 00:11Z.